Yared Berhanu Bodena



I am a Software Engineer and ML/AI Engineer with a strong foundation in Python and backend development, and hands-on experience in building real-world AI systems. I specialize in Machine Learning and Generative AI, including LLMs, RAG pipelines, and end-to-end AI applications using tools such as Hugging Face, LangChain, and OpenAI. I have experience designing scalable APIs, integrating AI models into production systems, and applying MLOps practices for deployment, monitoring, and iteration, with a strong focus on delivering reliable, high-quality AI solutions that create real business value.

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WORK EXPERIENCE

JU Inovation and Incubation Center Sep 2023 -January 2024

M L E n g i n e e r

- Developed an end-to-end AI assistant for document analysis, enabling semantic search and question-answering over large volumes of reports, filings, and earnings calls.
- Built a RAG pipeline with embeddings, vector search, and LLM integration (OpenAI & Hugging Face) to deliver precise, context-aware insights.
- Designed and deployed scalable APIs with FastAPI, enabling seamless integration with internal tools and dashboards for monitoring model performance.
- Achieved ~70% reduction in manual research time for financial analysts while improving accuracy of responses to complex queries.
- Utilized MLOps practices for deployment, monitoring, and iterative improvements, ensuring reliability in production-like environments.

EDUCATION

Jima University

Bachelor of Science in Software Engineering 2021-2025

PROJECTS

Intelligent Customer Support Chatbot 2022-2023

- Built an AI-powered support assistant for enterprises capable of handling complex customer queries using RAG pipelines and LLMs.
- Designed a document ingestion system to index knowledge bases, manuals, and FAQs, enabling context-aware responses for diverse support requests.
- Developed a scalable backend API using FastAPI and integrated with web and mobile frontends, ensuring real-time responsiveness.
- Implemented automated monitoring and analytics for response accuracy, user satisfaction, and system performance, applying MLOps principles for iterative improvement.
- Reduced support team workload by 60% and improved response consistency across channels, demonstrating measurable business impact.
- Tech Stack: Python | LangChain | OpenAI API | FAISS / Milvus | FastAPI | PostgreSQL | Docker | MLOps

Medical-Chatbot-With-LLMs-Langchain-Pinecone-Django

- Designed and developed an AI-powered medical assistant capable of answering patient queries, providing symptom guidance, and retrieving information from verified medical documents using LLMs and RAG pipelines.
- Built a document ingestion and vector search system with Pinecone, enabling context-aware responses from large-scale medical literature and clinical guidelines.
- Developed full-stack backend services with Django, exposing REST APIs for web and mobile frontends, and ensuring secure handling of sensitive medical data.
- Implemented monitoring, logging, and iterative improvements using MLOps practices, ensuring high accuracy and reliability for real-world deployment scenarios.

- Delivered a solution that streamlines patient interactions and provides instant, reliable access to medical knowledge while maintaining compliance with data privacy standards.
- Tech Stack: Python | Django | LangChain | OpenAI API | Pinecone | PostgreSQL | Docker | MLOps

Financial News & Sentiment Analysis Platform 2023 – 2024

- Built an AI system to analyze financial news, earnings reports, and market trends, generating actionable insights for traders and analysts.
- Implemented NLP pipelines with LLMs and RAG, combining semantic search with sentiment analysis to identify market signals in real-time.
- Developed a scalable backend API using FastAPI/Django, with vector search (FAISS/Pinecone) for fast retrieval of relevant news and historical trends.
- Applied MLOps practices including model monitoring, retraining, and performance logging to ensure high accuracy and low latency in production.
- Delivered a solution that improved decision-making speed by ~60% for analysts, while reducing manual news monitoring efforts.
- Tech Stack: Python | FastAPI/Django | LangChain | OpenAI API | FAISS / Pinecone | Docker | MLOps

SKILLS

- **Gen AI**: Large Language Models (LLM), LangChain, LangGraph, AgentOps, Multi-Agentic Systems, MCP,RAG,HugingFace,OpenAI
- **Software Engineering**: Software Architecture, Design Patterns, Event-Driven Architecture, Testing, Data Engineering
- **ML & Data Science**: PyTorch, TensorFlow, Computer Vision, CNN, Self-Supervised Learning, Medical Imaging, MONAI, Feature Engineering, Statistical analysis, Hypothesis testing, A/B testing, Autoencoders, Feature Store
- **Cloud & DevOps**: AWS, Kubernetes, Docker, Databricks, Delta Lake
- **Database**:FAISS, Pinecone,
- **Version Control:**Git, GitHub, GitLab, GitHub Actions, GitLab Pipelines

SOFT SKILLS

- **Communication**: Clear technical communication and effective stakeholder collaboration
- **Teamwork**: Collaborates well in cross-functional, agile team environments
- **Problem Solving**: Analytical, structured approach to complex technical challenges
- **Adaptability**: Quickly learns new tools and technologies in fast-evolving fields
- **Ownership & Time Management**: Dependable delivery of quality work against deadlines